| Responsible division : | Responsible unit : | Document type : | Confidentiality status : | |
|---|---|---|---|---|
| PGR | MLN/ENG | Requirement specification | Internal | BOMBARDIER |
| Prepared : 2018-05-15 G. Parisi | | Title : **System Function Req Spec Coupler V300ZEFIRO Italy** | Document state : Released | |
| Checked : 2018-05-17 F. Brughera | | | **3EST000225-7903** | |
| Approved : 2018-05-18 M. Brusco | | File name : 3EST000225-7903_Ben.docx | Revision : _B | Language : en / Pages : 1/38 |
| | | External Revision Date: External Approver: | Customer External Identity: | |
| | | Saved: 2018-05-18 18:00 | Sequence index : 1 | |

# Coupler-VF.CONCEPT

# SFRS for Zefiro-Europe

Generated from DOORS module /Platform.-.Zefiro-Europe/01.-.Requirements Breakdown Structure.-.Zefiro-Europe/2F.-.Vehicle-Functional Integration Areas.-.Zefiro-Europe/2F_02.-.Exterior Fittings-VF.-.Zefiro-Europe/2F_02.04.-.Coupler-VF.-.Zefiro-Europe/Coupler-VF.CONCEPT, baseline 2.3

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## Content

**1 Introduction 3**

**1.1 Train Configuration 3**

**1.2 Definitions and abbreviations 3**

**2 System Overview 4**

**2.1 System Configuration 4**

**3 Functional requirements 5**

**3.1 Coupling 5**
3.1.1 Function Design 5
3.1.2 Requirements 7

**3.2 Decoupling 11**
3.2.1 Function Design 11
3.2.2 Requirements 11

**3.3 Control of Coupler Cover 14**
3.3.1 Function Design 14
3.3.2 Requirements 15

**3.4 Coupler heating 21**
3.4.1 Function Design 21
3.4.2 Requirements 21

**4 Signal Interface 22**

**4.1 I/O Interface 22**
4.1.1 Supplier system 22
4.1.2 TCMS SW 22

**4.2 Bus Interface 24**

**5 References 25**

**6 Revision Log 26**

**6.1 Revision log 3EST000225-7903_A 26**

**6.2 Revision log 3EST000225-7903_B 36**

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

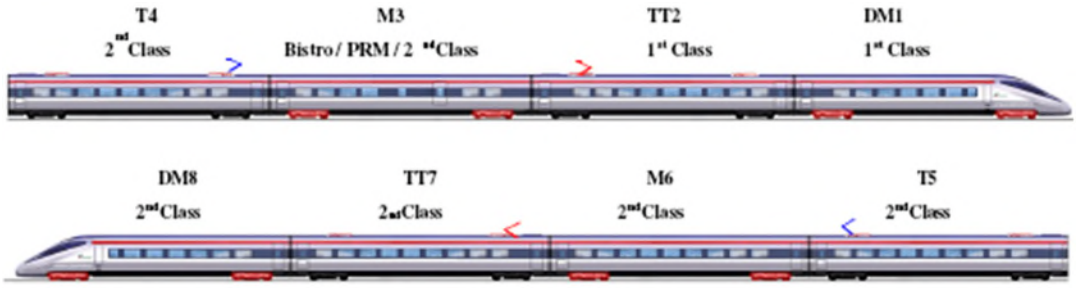
©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

# 1 Introduction

This document is a function specification for the control of the *Coupler* and *Coupler Covers* system for V300ZEFIRO for customer Trenitalia.

## 1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:




- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

## 1.2 Definitions and abbreviations

| | |
|---|---|
| CD | Circuit Diagram |
| EVENT | Diagnostics that will be stored in the Event database within TCMS SW. |
| HW HMI | Hardware Human Machine Interface, e.g. buttons, switches and indication lamps. |
| IDU | Interface Display Unit |
| IPB | Illuminated Push Button |
| TCMS | Train Control and Management System |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

# 2 System Overview

## 2.1 System Configuration

The SFRS module coupler consist of two separate systems:

**Coupler:** Connects two train sets with automatic coupler mechanically, electrically and pneumatically.

**Coupler covers:** Covers the coupler in single traction and opens front with automatic coupler for double traction.

Coupling sequence is activated by a control device on driver's desk as described in SFRS Conventional Train Control. The coupler covers can be opened via buttons on the IDU. Decoupling is activated by a control device on the driver’s desk. The coupler cover’s are closed by a push button on the IDU and automatically at a certain speed.

The train level function of coupling and decoupling are described in the SFRS conventional train control [2]. The requirements for the coupling control are described within this document.

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

# 3 Functional requirements

## 3.1 Coupling

### 3.1.1 Function Design

Coupling is a process to create a mechanical and electrical couple of two trains. By this, pneumatic and electric signals can be transferred between both parts. Coupling is also possible between a train and a loco, whereas no electrical coupling can be done in this case.

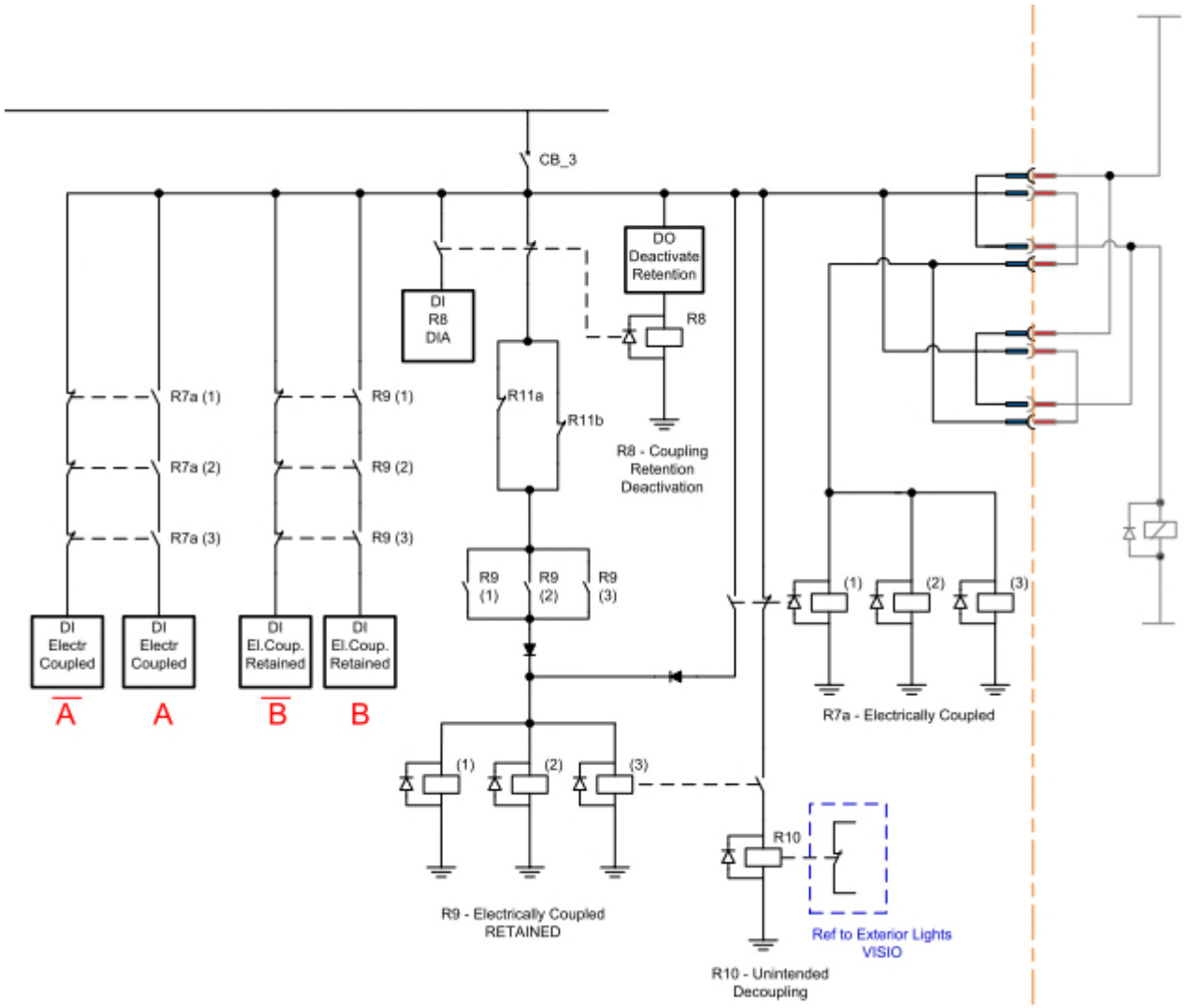
The coupling function requires the activation of the “Couple Speed Mode” for which all the conditions are described in SFRS conventional train control [2]. Herein, also the switch to access this mode is described. When activating this mode, the coupler cover in the front will open automatically. By this coupler mode also a speed limit is given which ensures the automatic coupling without damaging the coupler. Moreover, the coupler cover can be opened via the IDU below a certain speed.

Following signals are needed to define the status of coupling condition:
Mechanical coupled/uncoupled from mechanical switch
Electrical coupled indicated by “Signal A” which is high in case electrical coupler is fully connected - Relays R7a.
Electrical coupled indicated by “Signal B” which is high in case electrical coupler was once fully connected and not intended decoupled - Relays R9.
Signal B has a stored function and is used to detect unintended decoupling.

A detailed description of the coupler cover control can be found in chapter 3.3
Electrical connection of coupler relays

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.




### 3.1.1.1 Normal condition

In normal couple condition the couple sequence shows no failure related to coupler cover opening. Both mechanical and electrical couple conditions are aligned and Signal A and Signal B for electrical coupling are high.

### 3.1.1.2 Failure cases

1. Electrical coupled detected but not coupled
2. Electrical coupled but not detected
3. Mechanical coupled but mechanical coupled signal not set by coupler
4. Not mechanical coupled but mechanical coupled signal set by coupler
5. Implausible electrical coupling Signal A and B

### 3.1.1.3 Failure detection

1. When electrical is set and mechanical coupled not set at the same time an event shall be set
2. When electrical is not set and mechanical coupled is set at the same time an event shall be set
3. When mechanical coupled not set and electrical coupled is set at the same time an event shall be set
4. unintended decoupling is result. Electrical coupled Signal A and Signal B used for uncoupling detection
5. electrical coupled Signal A and Signal B used for coupling status detection


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

#### 3.1.1.4 Failure handling

1. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
2. No multiple traction possible. Only towing possible.
3. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
4. Emergency brake applied. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
5. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.

### 3.1.2 Requirements

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.46 |
| **Description** | The coupler shall automatically connect two trains together mechanically, pneumatically and electrically and make it possible to operate the whole train from the front-end cab. |
| **Safety level** | N/A |

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.42 |
| **Description** | Emergency brake shall automatically be applied in both trainparts in case of unintended decoupling. |
| **Safety level** | SIL 2 |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.360 | Automatic Coupled shall provide status of coupling device via microswitches installed on coupled head:<br>Mechanical Coupled (Switch S14)<br>Mechanical UNCoupled (Switch S15) | Derived Requirement | 4S_02.06.-.Couplers | | |
| 2F_02.04.Zefiro-Europe.CONCEPT.301 | The signal of the mechanical coupling switch (S14) and mechanical uncoupled switch (S15) shall be connected to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.302 | TCMS shall acquire following status<br>S14 = 1 -> Mechanically coupled switch set (acquired using K2 delayed opening relay) | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.303 | TCMS shall acquire following status<br>S15 = 1 -> Mechanically Uncoupled switch set. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

There are two signals “Signal A” and “Signal B” for defining the electrical coupled condition.
"Signal A" shall be derived from series of all R7a relays NO contacts
"Signal not-A" shall be derived from series of all R7a relays NC contacts

"Signal B" shall be derived from series of all R9 relays NO contacts
"Signal not-B" shall be derived from series of all R9 relays NC contacts

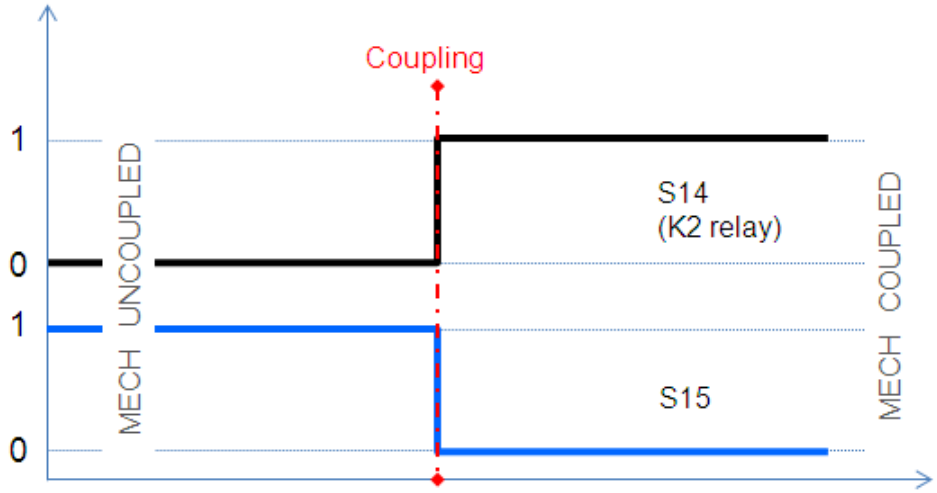
| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.304 | Relays detecting the electrical coupling connection (R7a) shall be installed.<br>The series of NO contacts of all R7a relays shall be connected to TCMS<br>The series of NC contacts of all R7a relays shall be connected to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

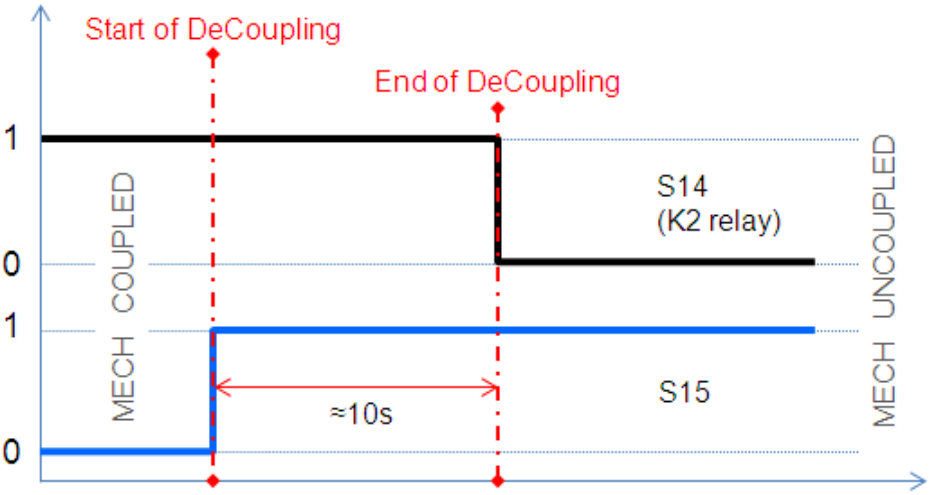
©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.305 | Relays retaining the electrical coupling connection (R9) shall be installed (relay status shall be mainained after the first connection untill a dedicated reset is done).<br>The series of NO contacts of all R9 relays shall be connected to TCMS<br>The series of NC contacts of all R9 relays shall be connected to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |

Mechanical Coupling sequence signals




Mechanical DeCoupling sequence signals





This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.307 | Following logic shall be used to define the Train couple condition:<br><br>S14=1 & S15=0 → Train Mechanically coupled (mc)<br><br>S14=0 & S15=1 → Train Mechanically Uncoupled (muc)<br><br>A=1 & B=1 → Train Electrically Coupled (ec)<br><br>A=0 & B=0 → Train Electrically Uncoupled (euc)<br><br>A=0 & B=1 → Unintended decoupling (T1)<br><br>(mc) & (ec) → Train coupled multiple operation (T3)<br><br>(mc) & (euc) → Train mechanically coupled only (T5)<br><br>(muc) & (euc) → Train uncoupled (T6) | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.207 | In case of Unintended decoupling (T1) an alarm shall be displayed to the driver | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.364 | For coupling conditions T3 and T5 an event shall be set | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.308 | Following logic shall be used to define the faulty status of couple conditions:<br><br>A=1 & B=0 → Faulty coupling retained logic (T2)<br><br>(muc) & (ec) → No mechanical coupling detected (only electrical) (T4) | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.309 | In case of train couple failure condition are set for more than 15s,<br>T2 - unexpected electrical Failure<br>T4 - unexpected electrical Failure<br>an event shall be generated for every case above | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.363 | In case mechanical coupling status cannot be detected<br>→ S14 = S15 for more than 15s (or both signals are not valid)<br>OR<br>Mechanical coupling status changes at speed (if v > 3km/h)<br><br>Following function shall be inhibited<br>– Coupler Covers Automatic closing if v>10 km/h (only in the cab where Coupling status cannot be detected)<br>– Remote decouplig that affects automatic coupler in the cab where Coupling status cannot be detected<br><br>An event shall be set to the driver | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.355 | "Signal A" and "Signal not-A" shall always have opposite values. In case of incoherence an event shall be set suggesting a maintenance check on coupling relays | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.356 | "Signal B" and "Signal not-B" shall always have opposite values. In case of incoherence an event shall be set suggesting a maintenance check on coupling relays | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.310 | Train coupled configuration shall be displayed in IDU menus | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.349 | After an "unintended decoupling" is detected, TCMS shall reset "Signal B" ("Electrically coupled retained status") by setting a DO, only when the train is in standstill and the driver has acknowledged the new train configuration | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.350 | The storage of the "unintended decoupling" signal shall be designed to be available at every possible state of the battery. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | |
| 2F_02.04.Zefiro-Europe.CONCEPT.96 | The electrical coupled connection shall be verified by dedicated pins in each of the electrical coupler heads to generate signal electrical coupled. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_02.04.Zefiro-Europe.CONCEPT.209 | When electrical heads loose contact the emergency brake loop actuator line shall be de-energised in both train sets. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## 3.2 Decoupling

### 3.2.1 Function Design

The driver is responsible for the operation of the decoupling command. After decoupling the coupler cover closes automatically at a certain speed further described in chapter 3.3. In case of manual cover close command the driver has to check that there is no obstruction interfering the closure of the coupler cover.

Decoupling is activated by a control device on driver's desk. Depending on the position of a rotary switch the local or remote coupler will be decoupled. The two decoupling magnet valves EVCD 02.14 and EVCD2 02.20 are in scope of brake system.

When decoupling order is received and the train has applied parking brakes the coupler will be decoupled pneumatically.

The supervision of the uncouple condition is described in the couple chapter 3.1.

#### 3.2.1.1 Failure condition

1. Decoupling order is given but coupler is not decoupling
2. Decoupling order is not given but coupler is decoupling

#### 3.2.1.2 Failure detection

1. The train coupler condition is supervised by electrical and mechanical switches, refer to 3.1
2. Detected by unintended decoupling, refer to 3.1

#### 3.2.1.3 Failure handling / effect

1. manual decoupling by authorised stuff. No impact on train operation.
2. emergency brake activated

### 3.2.2 Requirements

| **ID** | 2F_02.04.Zefiro-Europe.TRS.77 |
|---|---|
| **Description** | The driver shall be able to order the two trainsets to decouple by a control in the cab. |
| **Safety level** | |

| **ID** | 2F_02.04.Zefiro-Europe.TRS.78 |
|---|---|
| **Description** | Decoupling of two trainsets shall only be possible from the active cab. It shall be possible to disconnect the two trainsets in front of the active cab (if a cab in the middle of the 16 cars train is activated) or on the opposite side in remote (if a cab at the end of the 16 cars is activated and the disconnection is done in the middle). |
| **Safety level** | |

| **ID** | 2F_02.04.Zefiro-Europe.TRS.79 |
|---|---|
| **Description** | Decoupling shall only be possible if the train is standing still. |
| **Safety level** | |

| **ID** | 2F_02.04.Zefiro-Europe.TRS.49 |
|---|---|
| **Description** | Two trainsets shall be decoupled when a decoupling order from the cab is received. |
| **Safety level** | N/A |

For both, remote and local decoupling, the valves shall be energized for 10s.

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.312 | There shall be a decoupling control device on the driver's desk. | Derived Requirement | 4S_08.07.01-.Drivers Desk | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.313 | The different positions of the decoupling control device<br>• Local<br>• 0<br>• Romote<br>shall be connected to TCMS digital inputs | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.320 | When remote and local decoupling are requested both decoupling valves shall be energized. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits<br>4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.321 | To ensure a successful decoupling, the "mechanically coupled" relay shall be designed as a time relay maintaining its active state for 10 seconds. The relay shall be controlled by the coupler switch S14 "mechanically coupled" | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.171 | TCMS outputs for remote and local decoupling shall be connected to the two decoupling valves for coupler. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.163 | When remote de-coupling sequence is started TCMS shall energize the decoupling valves for coupler in the rear end of the train | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.388 | When local de-coupling sequence is started TCMS shall energize the decoupling valves for coupler in the rear end of the train | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.374 | When any de-coupling command is set (local or remote) TCMS shall check that both decoupling valves are correctly energized.<br>In case one or more feedback signals are not set an event shall be displayed (decoupling not possible) | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.375 | When there is NO de-coupling command set (local and remote) TCMS shall check that both decoupling valves are correctly de-energized.<br>In case one or more feedback signals are unduly set an event shall be displayed (isolate decoupling pipe to avoid unwanted decoupling) | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.347 | In case of decoupling commanded by TCMS the valves shall be energized for 10s. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.315 | It shall be possible to start the remote and local decoupling only if:<br>• The decoupling command comes from the active cab AND<br>• the Master Controller is in neutral position (coast) AND<br>• the train is at standstill AND<br>• Reverse direction is NOT selected AND<br>• min 75% of parking brakes are applied in each train of the consist individually | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.316 | An advisory event shall be set in case of remote decoupling is not allowd reporting the missing condition | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.317 | Decoupling shall be considered as successful if following conditions are true:<br>– Train was previously in a valid coupled status → (T3) or (T5)<br>– Train coupled condition becomes "uncoupled" → (T6) | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.318 | If the decoupling is commanded, Signal B (El coupled retained status) shall be restored de-energizing R9 relays | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.319 | There shall be the possibilty to reset the stored Signal B de-energizing R9 relays, by a TCMS output | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.80 |
| **Description** | After uncoupling, the decoupled train part shall apply emergency brake. In the part with active cab the new configuration of the security loops shall be acknowledged by the driver. |
| **Safety level** | |

Brake condition of a train with deactivated cab and not applied parking brake can be found in Brake SFRS.

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.81 |
| **Description** | Failure of the decoupling sequence shall be indicated to the driver. |
| **Safety level** | |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.201 | DELETED | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.323 | When decoupling is requested and "mechanically decoupled" signal is not present after 30s and Event shall be generated and shown on IDU | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

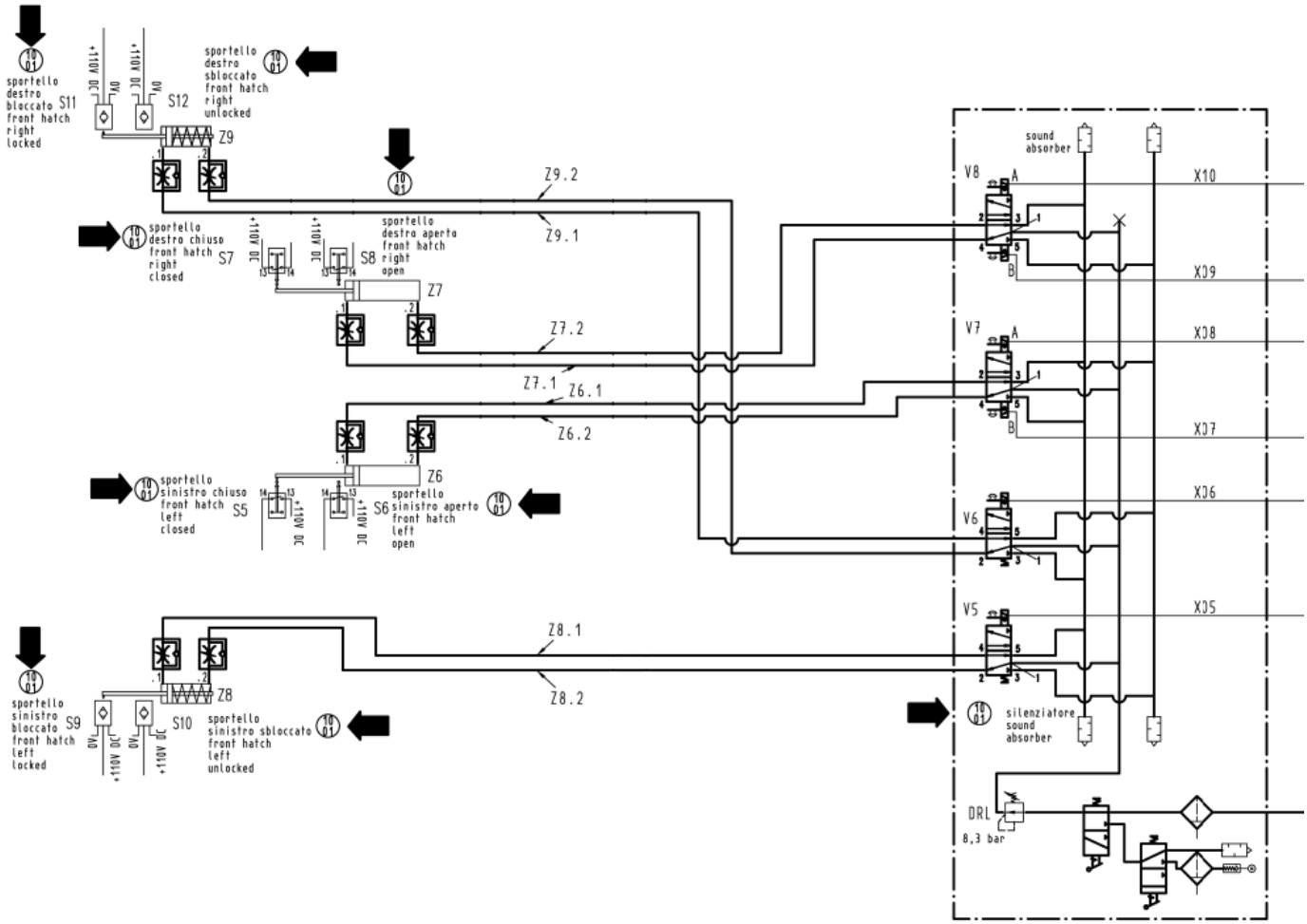
## 3.3 Control of Coupler Cover

### 3.3.1 Function Design

The coupler covers of any end of the train set can be opened/closed from an active cab. The opening of coupler cover can be done in two ways. On one hand, the coupler cover in the front is opened automatically after the activation of couple speed mode. On the other hand, the cover can be opened by using forcing it via IDU. This is possible only below a certain speed. The cover will start to close automatically after exceeding a speed of 10km/h. Moreover, the cover can be closed via IDU button. The driver then is responsible the check the coupler environment and has to ensure no obstruction can collide with the cover.

Control of coupling cover and supervision if of any failures in the opening or closing sequence are present done by separate magnet valves and final position switches. The opening/closing sequence, following explained using the example of opening, consists of three phases.

1. Both, the right and left cover hatch are commanded to be unlocked by energising the respective magnet valves. Final position switches detect the complete unlocking.
2. Both, the right and left cover hatch are commanded to be open by separate magnet valves operating a pressure cylinder. Final position switches detect the successfully opened hatch.
3. Following, both hatches are locked by de-energising the magnet valves. Again, the final position is detected by separate position switches.





This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

#### 3.3.1.1 Normal conditions

In normal condition the coupler cover in the front of the driving direction opens as soon as the Coupler Speed Mode is activated. The cover is locked after the open position is detected. For closing the cover, first the hatches are unlocked and are closed then completely. After closure is detected, the hatches are locked again.

#### 3.3.1.2 Failure cases

1. coupler covers do not open automatically after activation of coupler speed mode
2. one or both coupler cover are not opening / closing when pressing IDU coupler cover control buttons
3. coupler covers closing without order
4. coupler covers are closed/ opened but not locked
5. Implausible indication of locked/unlocked or open/close
6. Coupler covers remain open at speed greater than 50km/h

#### 3.3.1.3 Failure detection

1. Coupler status supervised by final position sensor and indicated to the driver
2. Coupler status supervised by final position sensor and indicated to the driver
3. Coupler status supervised by final position sensor and indicated to the driver
4. Coupler status (including locked/unlocked) supervised by final position sensor and indicated to the driver
5. if there is a implausible indication, a failure event will be created
6. Coupler status supervised by final position sensor and indicated to the driver

#### 3.3.1.4 Failure handling

1. Opening of cover can be command by IDU as a second path. If this won't work either, the cover can be opened manually.
2. Covers must be opened manually.
3. Opening must be commanded again. If failure remains present, cover must be opened manually.
4. Pneumatic path must be vented. Spring loaded cover lock will close automatically.
5. If status of coupler cover is unclear, it must be controlled manually
6. Speed limit of 140km/h is set.

### 3.3.2 Requirements

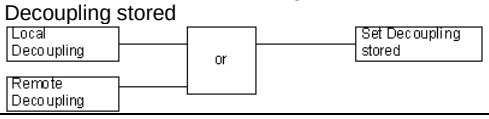
| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.66 |
| **Description** | The driver shall be able to order the coupler cover to close by a control in the drivers desk after the train has been decoupled. |
| **Safety level** | N/A |

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.63 |
| **Description** | The opening of the coupler covers shall be interlocked at speeds >10 km/h. |
| **Safety level** | TBD |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.369 | TCMS shall acquire following signals to define coupler covers status<br><br>Sensor S5 → LEFT hatch CLOSED<br>Sensor S6 → LEFT hatch OPEN<br>Sensor S7 → RIGHT hatch CLOSED<br>Sensor S8 → RIGHT hatch OPEN<br>Sensor S9 → LEFT hatch LOCKED<br>Sensor S10 → LEFT hatch UNLOCKED<br>Sensor S11 → RIGHT hatch LOCKED<br>Sensor S12 → RIGHT hatch UNLOCKED | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

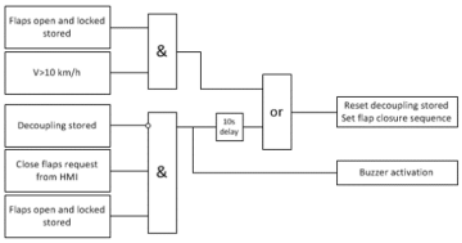
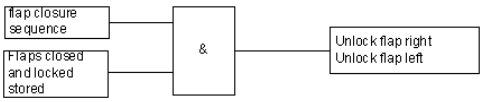
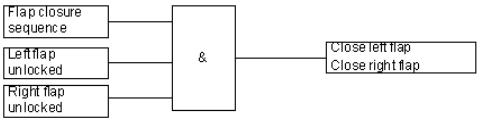
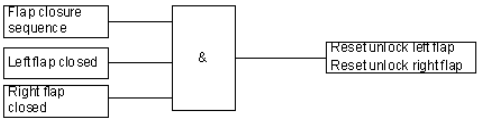
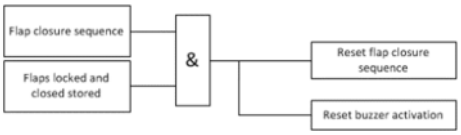
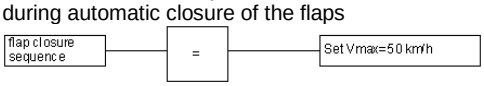

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.370 | TCMS shall command control valves for front cover opening/closing/locking<br><br>V7A/V8A → Front hatches CLOSING command<br>V7B/V8B → Front hatches OPENING command<br>V5/V6 → Front hatches UNLOCK command | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.338 | The status of **OPEN/CLOSED** coupler flaps shall be stored within TCMS according following logic:<br><br>S5=0 & S6=1 → **LEFT flap OPEN**<br>S5=1 & S6=0 → **LEFT flap CLOSED**<br><br>S5 = S6 or<br>any S5, S6 not valid → **LEFT flap position FAULTY**<br><br>S7=0 & S8=1 → **RIGHT flap OPEN**<br>S7=1 & S8=0 → **RIGHT flap CLOSED**<br><br>S7 = S8 or<br>any S7, S8 not valid → **RIGHT flap position FAULTY** | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.357 | The status of **UNLOCKED/LOCKED** coupler flaps shall be stored within TCMS according following logic:<br><br>S9=1 & S10=0 → **LEFT flap LOCKED**<br>S9=0 & S10=1 → **LEFT flap UNLOCKED**<br><br>S9 = S10 or<br>any S9, S10 not valid → **LEFT flap locking FAULTY**<br><br>S11=1 & S12=0 → **RIGHT flap LOCKED**<br>S11=0 & S12=1 → **RIGHT flap UNLOCKED**<br><br>S11 = S12 or<br>any S11, S12 not valid → **RIGHT flap locking FAULTY** | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.373 | In case any cover is detected Faulty (position faulty OR locking faulty) for more than 15s,<br>– automatic closing shall be inhibited | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.371 | In case any cover is detected Faulty (position faulty OR locking faulty) for more than 15s,<br>a warning shall be set to the driver to check and override cover position using IDU commands | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.332 | If local decoupling or remote decoupling is selected TCMS shall set the signal Decoupling stored<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

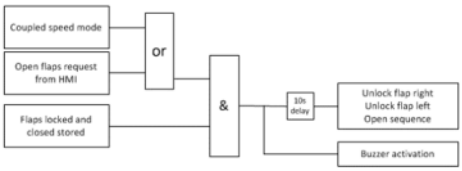
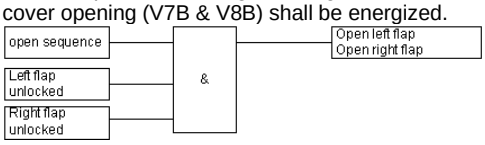
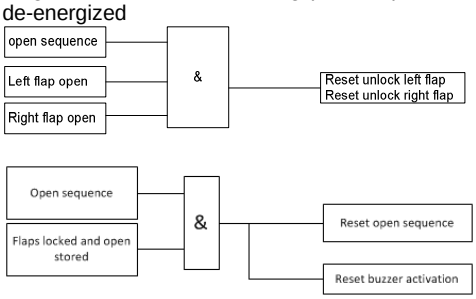
©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.182 | TCMS shall start the coupler covers closing sequence when following conditions are set:<br>– Train speed is above 10 km/h<br>– The train has just decoupled and accelerates from that condition<br>– Local automatic coupler status is NOT FAULTY<br>– Local automatic coupler status is MECHANICALLY UNCOUPLED<br>– Local cover status is OPEN and LOCKED (both left and right flap)<br>– Local cover status is NOT FAULTY (both left and right flap)<br>– NO UNINTENDED DE-COUPLING has occurred<br><br>The closure is requested by soft key and blocked in case the train has decoupled and not jet moved away from the decoupled train<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.335 | The closure sequence shall unlock the flaps,<br>  <br>close the flaps<br>  <br>and lock the flaps when closed<br>  <br>and reset the flap closure and buzzer activation when finished<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.333 | TCMS shall set max speed of the train to 50km/h during automatic closure of the flaps<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.47 |
| **Description** | The driver shall be able to order the coupler covers to open by a control in the drivers desk. |
| **Safety level** | SIL 0 |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.63 | The IDU shall provide a status menu for coupler and coupler covers to show the actual condition in front or rear car end:<br>– Coupler cover opened/closed<br>– Coupler cover locked/unlocked | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.117 | For opening/closing the coupler covers at standstill there shall be following buttons on the IDU in the actived cab:<br>for the driver:<br>– Open/close coupler cover at front end of train<br>– Open/close coupler cover at rear end of train<br><br>– Set flaps locked and closed stored local/ remote (signals from position/locking sensor will be bypassed)<br>– Set flaps open and locked stored local/ remote (signals from position/locking sensor will be bypassed)<br><br>(It shall be prevented by TCMS software that the signals "flaps open and locked stored" and "flaps closed and locked stored" can be active at the same time. Whenever one is selected, the other shall be reset) | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.66 | The manual (via IDU) opening/ closing of the front cover shall not be allowed at train speed above 10km/h | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.324 | When opening command is sent by TCMS or by IDU button, first the magnet valves for cover locking (V6 & V5) are energised.<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.325 | After completed unlocking, the magnet valves for cover opening (V7B & V8B) shall be energized.<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.326 | When coupler cover are detected as open, the magnet valves for cover locking (V6 & V5) shall be de-energized<br>  | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.166 | TCMS output shall be connected to both unlock coupler cover magnet valve. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.167 | TCMS output shall be connected to both coupler cover open/close magnet valves. Each has one open and one close solenoid. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.170 | Final position switches for left and right coupler cover shall be connected to TCMS input:<br>– Coupler covers closed<br>– Coupler covers opened<br>– Coupler covers unlocked<br>– Coupler covers locked | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.119 | In case of Automatic Coupler status is detected ad "MECHANICALLY COUPLED" (mc), related front hatches shall be prevented from closing | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.51 |
| **Description** | Failure of opening or closing of the coupler cover shall be indicated to the driver. |
| **Safety level** | N/A |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.118 | When coupler covers are not opening or closing on order an event shall be set | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.161 | When coupler covers is not locked or not unlocked on order an event shall be set. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.262 | When coupler cover status change without order, an event shall be set | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.298 | Speed Limitation shall be set to 250km/h (parameter to be defined during commissioning) if following condition are verified:<br>– Any FRONT Flap OPEN (or in Open&Locked overridden position)<br>AND<br>– Automatic coupler status is MECHANICALLY UNCOUPLED<br>AND<br>– Coupler Covers NOT in Closed&Locked overridden position<br>AND<br>– NO UNINTENDED DE-COUPLING has occurred | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.346 | In case of open covers, over 50 Km/h a alarm will be displayed on the IDU | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.340 | DELETED | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.376 | When a manual (via IDU) opening of the front/rear flaps is requested the related buzzer shall be commanded by the TCMS 5 seconds before "open sequence" is set | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.384 | When a manual (via IDU) closing of the front/rear flaps is requested the related buzzer shall be commanded by the TCMS 5 seconds before "closure sequence" is set | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | | | |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.385 | Buzzer activation command shall be reset if at least one of the following conditions is verified:<br>– Open sequence is completed<br>OR<br>– Closure sequence is completed<br>OR<br>– Stop command control valves for front cover opening/closing/locking | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.386 | Two redundant TCMS digital outputs shall be connected to the buzzer | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.387 | As long as buzzer activation commad is set, an "other" event shall be set to mantainer | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## 3.4 Coupler heating

### 3.4.1 Function Design

Heating of coupler is activated in winter conditions at a temperature below 8°C.

When mechanical coupled is detected the heating of each connected coupler is reduced to save energy.

For outside temperature signal the sensor of air condition is used to control coupler heating.

#### 3.4.1.1 Normal condition

In uncoupled condition there are normally 5 heating elements switched on. In coupled condition two of 5 heating elements are switched off of each coupler. The respective heating elements are switch on below 1°C. They are switched off above 8°C. This behaviour is valid for both, coupled and uncoupled condition.

#### 3.4.1.2 Failure case

1. Heating is not activated when ordered
2. Heating is activated without order

#### 3.4.1.3 Failure detection

1. no failure detection possible
2. no failure detection possible

#### 3.4.1.4 failure handling

1. no influence on train operation
2. no influence on train operation

### 3.4.2 Requirements

| | |
|---|---|
| **ID** | 2F_02.04.Zefiro-Europe.TRS.68 |
| **Description** | Heating cartridge in coupler shall be activated by TCMS if the outside temperature is < 1º C. When heating is already activated it shall be deactivated if the outside temperature is > 8º C. |
| **Safety level** | N/A |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.101 | Heating cartridge in coupler shall be activated by TCMS if the outside temperature is < 1º C. When heating is already activated it shall be deactivated if the outside temperature is > 8º C. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_02.04.Zefiro-Europe.CONCEPT.155 | TCMS output shall be connected to coupler heating cartridge to allow both heating couples to be controlled indipendendly. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.217 | When signal mechanical coupled is set 2 of 5 heating elements shall be switched off. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_02.04.Zefiro-Europe.CONCEPT.330 | When decoupled all heating elements shall be switched on | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

# 4 Signal Interface

## 4.1 I/O Interface

### 4.1.1 Supplier system

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.124 | Close coupler covers right (magnet valve V8A) | L | 2F_02.04.Zefiro-Europe.CONCEPT.117 |
| 2F_02.04.Zefiro-Europe.CONCEPT.172 | Open coupler covers right (magnet valve V8B) | L | 2F_02.04.Zefiro-Europe.CONCEPT.117 |
| 2F_02.04.Zefiro-Europe.CONCEPT.240 | Close coupler covers left (magnet valve V7A) | | |
| 2F_02.04.Zefiro-Europe.CONCEPT.241 | Open coupler covers left (magnet valve V7B) | | |
| 2F_02.04.Zefiro-Europe.CONCEPT.173 | Unlock coupler covers right (magnet valve V6) | L | 2F_02.04.Zefiro-Europe.CONCEPT.117 |
| 2F_02.04.Zefiro-Europe.CONCEPT.242 | Unlock coupler covers left (magnet valve V5) | | 2F_02.04.Zefiro-Europe.CONCEPT.117<br>2F_02.04.Zefiro-Europe.CONCEPT.166 |
| 2F_02.04.Zefiro-Europe.CONCEPT.216 | Coupler heating | L | 2F_02.04.Zefiro-Europe.CONCEPT.101<br>2F_02.04.Zefiro-Europe.CONCEPT.155 |
| 2F_02.04.Zefiro-Europe.CONCEPT.218 | Coupler heating when coupled | L | 2F_02.04.Zefiro-Europe.CONCEPT.217 |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.187 | Mechanical coupled | L | |
| 2F_02.04.Zefiro-Europe.CONCEPT.194 | Mechanical uncoupled | L | |

### 4.1.2 TCMS SW

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.30 | Electrical coupled | L | |
| 2F_02.04.Zefiro-Europe.CONCEPT.125 | Electrical coupled stored | L | |
| 2F_02.04.Zefiro-Europe.CONCEPT.188 | Mechanical coupled | L | 2F_02.04.Zefiro-Europe.CONCEPT.201 |
| 2F_02.04.Zefiro-Europe.CONCEPT.195 | Mechanical uncoupled | L | 2F_02.04.Zefiro-Europe.CONCEPT.201 |
| 2F_02.04.Zefiro-Europe.CONCEPT.126 | Coupler cover status open right S8 | L | 2F_02.04.Zefiro-Europe.CONCEPT.118<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |

This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.243 | Coupler cover status open left S6 | | 2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.244 | Coupler cover status close right S7 | | 2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.127 | Coupler cover status close left S5 | L | 2F_02.04.Zefiro-Europe.CONCEPT.118<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.177 | Coupler cover status unlock right S12 | L | 2F_02.04.Zefiro-Europe.CONCEPT.161<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.248 | Coupler cover status unlock left S10 | | 2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.178 | Coupler cover status lock right S11 | L | 2F_02.04.Zefiro-Europe.CONCEPT.161<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.249 | Coupler cover status lock left S9 | | 2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.189 | Heating active | L | |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.31 | Decoupling valve V4 (Brake scope 02.14/02) | L | 2F_02.04.Zefiro-Europe.CONCEPT.163<br>2F_02.04.Zefiro-Europe.CONCEPT.171 |
| 2F_02.04.Zefiro-Europe.CONCEPT.250 | Decoupling valve (stop valve) V0 (Brake scope 02.14/01) | | 2F_02.04.Zefiro-Europe.CONCEPT.171 |
| 2F_02.04.Zefiro-Europe.CONCEPT.210 | Electrical coupled stored | L | |
| 2F_02.04.Zefiro-Europe.CONCEPT.131 | Close coupler covers right (magnet valve V8A) | L | 2F_02.04.Zefiro-Europe.CONCEPT.117<br>2F_02.04.Zefiro-Europe.CONCEPT.167<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.251 | Open coupler covers right (magnet valve V8B) | | 2F_02.04.Zefiro-Europe.CONCEPT.167<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.132 | Close coupler covers left (magnet valve V7A) | L | 2F_02.04.Zefiro-Europe.CONCEPT.117<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.252 | Open coupler covers left (magnet valve V7B) | | 2F_02.04.Zefiro-Europe.CONCEPT.167<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.180 | Unlock coupler covers right (magnet valve V6) | L | 2F_02.04.Zefiro-Europe.CONCEPT.161<br>2F_02.04.Zefiro-Europe.CONCEPT.166<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.253 | Unlock coupler covers left (magnet valve V5) | | 2F_02.04.Zefiro-Europe.CONCEPT.166<br>2F_02.04.Zefiro-Europe.CONCEPT.262 |
| 2F_02.04.Zefiro-Europe.CONCEPT.133 | Activate/deactivate heating of coupler | L | 2F_02.04.Zefiro-Europe.CONCEPT.101 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## 4.2 Bus Interface

Supplier to TCMS SW:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.32 | Not applicable | | |

TCMS SW to Supplier:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_02.04.Zefiro-Europe.CONCEPT.33 | Not applicable | | |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## 5 References


[1] 3EST000225-7910 SFRS Brake control

[2] 3EST000225-7916 SFRS Conventional train control
[3] PXYZ Circuit Diagram Coupler (Preliminary, for internal use only)
[4] P 422 Circuit Diagram Coupler Covers (Preliminary 2011.09.15)



This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

# 6 Revision Log

## 6.1 Revision log 3EST000225-7903_A

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.1 | [2F_02.04.Zefiro-Europe.CONCEPT.161]: | Speed limit | There should be a speed limit if the front flaps cannot be closed or locked.<br><br>Proposed change:<br>CONCEPT.161<br>If coupler covers are not locked or not unlocked on order an event shall be set. If the coupler cover is not locked when the close command is given there shall be a speed limit. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.2 | [2F_02.04.Zefiro-Europe.CONCEPT.153]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.160]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.162]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.221]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.223]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.264]: | Failure modes | Failure modes are missing and need to be inserted in the SFRS. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.3 | [2F_02.04.Zefiro-Europe.CONCEPT.63]: | IDU Interface | IDU interface changed for coupler covers<br><br>2F_02.05.Zefiro-Europe.CONCEPT.63:<br>The IDU shall provide a status menu for coupler and coupler covers to show the actual condition from coupler covers:<br>– Coupler cover opened and locked front/rear<br>– Coupler cover closed and locked front/rear |
| 2F_02.04.ISSUES.-.Zefiro-Europe.4 | [2F_02.04.Zefiro-Europe.CONCEPT.124]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.172]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.173]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.240]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.241]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.242]: | I/O Interface system | System signal interfaces changed based on last supplier input |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.5 | [2F_02.04.Zefiro-Europe.CONCEPT.126]: [2F_02.04.Zefiro-Europe.CONCEPT.127]: [2F_02.04.Zefiro-Europe.CONCEPT.177]: [2F_02.04.Zefiro-Europe.CONCEPT.178]: [2F_02.04.Zefiro-Europe.CONCEPT.243]: [2F_02.04.Zefiro-Europe.CONCEPT.244]: [2F_02.04.Zefiro-Europe.CONCEPT.248]: [2F_02.04.Zefiro-Europe.CONCEPT.249]: | TCMS interface system input | TCMS signal interfaces changed based on last supplier input |
| 2F_02.04.ISSUES.-.Zefiro-Europe.6 | [2F_02.04.Zefiro-Europe.CONCEPT.131]: [2F_02.04.Zefiro-Europe.CONCEPT.132]: [2F_02.04.Zefiro-Europe.CONCEPT.180]: [2F_02.04.Zefiro-Europe.CONCEPT.250]: [2F_02.04.Zefiro-Europe.CONCEPT.251]: [2F_02.04.Zefiro-Europe.CONCEPT.252]: [2F_02.04.Zefiro-Europe.CONCEPT.253]: [2F_02.04.Zefiro-Europe.CONCEPT.31]: | TCMS interface system output | TCMS signal interfaces changed based on last supplier input |
| 2F_02.04.ISSUES.-.Zefiro-Europe.7 | [2F_02.04.Zefiro-Europe.CONCEPT.254]: [2F_02.04.Zefiro-Europe.CONCEPT.255]: | Add references | Missing supplier interfaces |
| 2F_02.04.ISSUES.-.Zefiro-Europe.8 | [2F_02.04.Zefiro-Europe.TRS.77]: [2F_02.04.Zefiro-Europe.TRS.78]: [2F_02.04.Zefiro-Europe.TRS.79]: [2F_02.04.Zefiro-Europe.TRS.80]: [2F_02.04.Zefiro-Europe.TRS.81]: | Move of decoupling from 2F CTC to 2F Coupler | The part of decoupling that was covered in SFRS CTC has been moved to Coupler SFRS.<br><br>Decoupling is initiated by the driver in the activated cab. The decoupling can be done for this cab or remote for the other cab, i.e in the middle of the train. There are two separate illuminated pushbuttons on the driver's desk for this, one to decouple in front of this cab and one to decouple in remote. To avoid that the driver is unaware of a decoupling performed in remote, e.g. that the button has been pressed by mistake, the driver has to acknowledge the new configuration before the emergency brake is released. The illuminated push button gives indication of the decoupling sequence to the driver. When decoupling is completed it is possible to close the coupler covers using a soft button on the IDU also for the decoupled train set<br><br>Description of acknowledge:<br>A principle function that driver has acknowledged the uncoupling(reset) shall be implemented . The |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| | | | reset function can be done for uncoupling in both ends from the activated cab.<br>The function makes keeps the coupling relay in coupled position so the EM-brake loop is open until<br>the driver do a reset with the reset decoupling function from active cab.<br>Note! An MR is needed for the acknowledge function because it uses the safe TCMS platform.<br><br>Decouple can only be done by the driver when the train is standing still. Unwanted train separation, for other reason than initiated by the driver is covered in chapter unwanted decoupling in the SFRS. See issue 9.<br><br>Old TRS.requirements from 2F_09.03 CTC (Charlotta)<br>ID: 2F_09.03.Zefiro-Europe.TRS.317<br>Description: The decoupling of two trainsets shall be possible in remote, even from the cab opposite the coupling side.<br>SIL: N/A<br><br>ID: 2F_09.03.Zefiro-Europe.TRS.316<br>Description: The driver shall be able to order the two trainsets to decouple by a control in the cab<br>SIL: N/A<br><br>ID: 2F_09.03.Zefiro-Europe.TRS.313<br>Description: Failure of the decoupling sequence shall be indicated to the driver.<br>SIL: N/A<br><br>ID: 2F_09.03.Zefiro-Europe.TRS.114<br>Description: Uncoupling of two trains shall only be possible from the active cab. The disconnection may be performed in front of the active cab (if a cab in the middle of the 16 cars train is activated) or on the opposite side in front of a not activated cab.<br>SIL: N/A<br><br>New TRS.requirements in 2F_02.04 (Charlotta)<br>ID: 2F_02-04.Zefiro-Europe.TRS.77<br>The driver shall be able to order the two trainsets to decouple by a control in the cab.<br><br>ID: 2F_02-04.Zefiro-Europe.TRS.78<br>Decoupling of two trainsets shall only be possible from the active cab. It shall be possible to disconnect the two trainsets in front of the active cab (if a cab in the middle of the 16 cars train is activated) or on the opposite side in remote (if a cab at the end of the 16 cars is activated and the disconnection is done in the middle).<br><br>Old derived requirements from 2F_09.03 CTC (Daniel/Tomas) |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| | | | |
|---|---|---|---|
| **Revision:** | **_A** | | |
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| | | | 2F_09.03.Zefiro-Europe.CONCEPT.469<br>Uncoupling shall be performed by giving a demand signal to the coupler control if:<br>– train speed is zero (0)<br>– uncoupling is demanded by the uncoupling button in the active cab<br>4S_09.03.05.-.TCMS Software<br><br>2F_09.03.Zefiro-Europe.CONCEPT.623<br>By pressing the "decoupling" button, the coupler shall be disconnected. The driver can select "decoupling front" or "decoupling rear".<br>TBD: two buttons needed? In towing more than 2 x 8 cars trains can be connected then it should be distinguished where to decouple.<br>4S_09.03.05.-.TCMS Software<br><br>2F_09.03.Zefiro-Europe.CONCEPT.624<br>Two buttons or one two position switch shall be provided in the cab:<br>– uncoupling demand in front<br>– uncoupling demand at rear<br>4S_08.07.01.-.Drivers Desk<br><br>2F_09.03.Zefiro-Europe.CONCEPT.679<br>TCMS shall control lamp in IPB "Decouple" in following order:<br>– Coupler cover close = not illuminated<br>– Coupler cover open and not coupled = illuminated<br>– Coupling sequence ongoing = flashing<br><br>Old TRS.requirement from 2F_09.03 CTC (Charlotta)<br>ID: 2F_09.03.Zefiro-Europe.TRS.318<br>Decoupling shall only be possible if the train is standing still.<br><br>New TRS.requirement (Charlotta)<br>ID: 2F_02.04.Zefiro-Europe.TRS.79<br>Description: Decoupling shall only be possible if the train is standing still.<br><br>Old derived requirements from 2F_09.03 CTC (Daniel/Tomas)<br>2F_09.03.Zefiro-Europe.CONCEPT.638<br>The decoupling command from the switch in the cab shall be interlocked with the zero speed wire to ensure that the train will standstill at if a decoupling command is sent to the coupler.<br>4S_09.02.09.-.Conventional Train Control – Circuits<br>SIL 2<br><br>Old TRS.requirement from 2F_09.03 CTC (Charlotta)<br>ID: 2F_09.03.Zefiro-Europe.TRS.233 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| | | | Description: After uncoupling, the decoupled train part shall apply emergency brake. In the part with active cab the new configuration of the security loops shall be acknowledged by the driver.<br>SIL: SIL 1<br><br>New TRS.requirement (Charlotta)<br>ID: 2F_02.04.Zefiro-Europe.TRS.80<br>After uncoupling, the decoupled train part shall apply emergency brake. In the part with active cab the new configuration of the security loops shall be acknowledged by the driver.<br><br>Old derived requirements from 2F_09.03 CTC (Daniel/Tomas)<br>2F_09.03.Zefiro-Europe.CONCEPT.472<br>The uncoupled train without active cab shall have emergency brake applied as long as no cab is activated<br>4S_09.02.09.-.Conventional Train Control – Circuits<br>SIL 1<br><br>2F_09.03.Zefiro-Europe.CONCEPT.671<br>Traction safe trainline shall be deenergised at decoupling until the driver has acknowledged the decoupling.<br>4S_09.02.09.-.Conventional Train Control – Circuits<br>SIL 1<br><br>2F_09.03.Zefiro-Europe.CONCEPT.628<br>The new configuration or WTB and security loops after decoupling shall be acknowledged by the driver on the IDU.<br>4S_09.03.05.-.TCMS Software<br>IDU<br>SIL 0<br><br>2F_09.03.Zefiro-Europe.CONCEPT.642<br>Traction block shall be enforced as long as the driver has not acknowledged the configuration.<br>4S_09.03.05.-.TCMS Software<br>SIL 0<br><br>Old TRS.requirement from 2F_09.03 CTC (Charlotta)<br>ID: 2F_09.03.Zefiro-Europe.TRS.313<br>Description: Failure of the decoupling sequence shall be indicated to the driver.<br>SIL: N/A<br><br>New TRS.requirement (Charlotta)<br>ID: 2F_02.04.Zefiro-Europe.TRS.81<br>Failure of the decoupling sequence shall be indicated to the driver.<br><br>Old derived requirements from 2F_09.03 CTC (Daniel/Tomas) |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| | | | 2F_09.03.Zefiro-Europe.CONCEPT.630<br>An event shall be indicated if the decoupling sequence is not performed within specified time window.<br>4S_09.03.05.-.TCMS Software<br>EVENT |
| 2F_02.04.ISSUES.-.Zefiro-Europe.10 | [2F_02.04.Zefiro-Europe.CONCEPT.262]: | Cover status change | the covers must only open/close after an order by the driver. If the status change without order an malfunction occurs |
| 2F_02.04.ISSUES.-.Zefiro-Europe.13 | [2F_02.04.Zefiro-Europe.CONCEPT.298]: | Flap closure during driving | For practical control of coupler covers it is either required to have an automatic closure at speed 15km/h or to have the possibility to open/close the covers via IDU. For the second option it is required to have accessibility to the respective IDU menu at speed up to 15km/h. Above that speed limit the manual control shall not be possible. Moreover, the driver shall be notified in case cover remain open at speed above 15km/h.<br><br>That implies a speed limitation in both cases during closure at speed 20km/h.<br><br>In case of automatic activation there shall be an IDU button that allows bypassing. Hereby, running with opened covers at speed higher than 20km/h is possible. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.14 | [2F_02.04.Zefiro-Europe.CONCEPT.298]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.346]: | Speed limitation in case of open covers | In case of open covers the speed shall be limited to 140 km/h (project decision required) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.15 | [2F_02.04.Zefiro-Europe.CONCEPT.101]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.290]: | Change of activation and deactivation temperatures of coupler heating function | Temperature thresholds for the activation and de-activation of the heating shall be set to 5°C(activation) and 8°C(de-activation) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.16 | [2F_02.04.Zefiro-Europe.CONCEPT.321]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.347]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.348]: | Deactivation of decoupling by time relay | To ensure a successful decoupling, the "mechanically coupled" relay shall be designed as a time relay maintaining its active state for 10 seconds. The relay shall be controlled by the coupler switch "mechanically coupled" |
| 2F_02.04.ISSUES.-.Zefiro-Europe.17 | [2F_02.04.Zefiro-Europe.CONCEPT.350]: | Availability of unintended decoupling signal | The storage of the "unintended decoupling" signal shall be designed to be available at every possible state of the battery. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.18 | [2F_02.04.Zefiro-Europe.CONCEPT.349]: | Removal of train configuration acknowledgement function | train configuration acknowledge:<br>COUPLER requirement 343, 344 deleted and moved to CTC SFRS concept.1559<br><br>2F_02.04.Zefiro-Europe.CONCEPT.349 added:<br>reset of unintended decoupling signal (Signal B) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.19 | [2F_02.04.Zefiro-Europe.CONCEPT.298]: | Change of requirement text | Remove the word "Serious" from the requirement to avoid misunderstanding |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.21 | [2F_02.04.Zefiro-Europe.CONCEPT.163]: [2F_02.04.Zefiro-Europe.CONCEPT.171]: [2F_02.04.Zefiro-Europe.CONCEPT.207]: [2F_02.04.Zefiro-Europe.CONCEPT.301]: [2F_02.04.Zefiro-Europe.CONCEPT.302]: [2F_02.04.Zefiro-Europe.CONCEPT.303]: [2F_02.04.Zefiro-Europe.CONCEPT.304]: [2F_02.04.Zefiro-Europe.CONCEPT.305]: [2F_02.04.Zefiro-Europe.CONCEPT.306]: [2F_02.04.Zefiro-Europe.CONCEPT.307]: [2F_02.04.Zefiro-Europe.CONCEPT.308]: [2F_02.04.Zefiro-Europe.CONCEPT.309]: [2F_02.04.Zefiro-Europe.CONCEPT.310]: [2F_02.04.Zefiro-Europe.CONCEPT.311]: [2F_02.04.Zefiro-Europe.CONCEPT.312]: [2F_02.04.Zefiro-Europe.CONCEPT.313]: [2F_02.04.Zefiro-Europe.CONCEPT.315]: [2F_02.04.Zefiro-Europe.CONCEPT.316]: [2F_02.04.Zefiro-Europe.CONCEPT.317]: [2F_02.04.Zefiro-Europe.CONCEPT.318]: [2F_02.04.Zefiro-Europe.CONCEPT.319]: [2F_02.04.Zefiro-Europe.CONCEPT.320]: [2F_02.04.Zefiro-Europe.CONCEPT.321]: [2F_02.04.Zefiro-Europe.CONCEPT.323]: [2F_02.04.Zefiro-Europe.CONCEPT.347]: [2F_02.04.Zefiro-Europe.CONCEPT.349]: [2F_02.04.Zefiro-Europe.CONCEPT.350]: | Change of functional structure – coupling/ unintended decoupling | The Chapters Coupling and Unintended decoupling have been merged because the identification logic for both states of the system follows the same logic.<br><br>An electrical signal is send via the electrical coupler and back to the origin vehicle to be stored if the connection is once achieved (wired solution to create the signal electrically coupled stored)<br>A second signal is checked all the time (wired solution to create the signal electrically coupled)<br><br>These to signals are evaluated together to determine the status of the vehicle which can be coupled, uncoupled or unintended decoupled<br><br>For a more detail the signals of the sensors which check the physical position of the mechanical coupler are used. Those sensors detect the states mechanically coupled and mechanically decoupled. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.22 | [2F_02.04.Zefiro-Europe.CONCEPT.330]: | Heating function – Update after Visio review | Added requirement to switch on all heating parts when train is decoupled according VISIO |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.23 | [2F_02.04.Zefiro-Europe.CONCEPT.118]: [2F_02.04.Zefiro-Europe.CONCEPT.119]: [2F_02.04.Zefiro-Europe.CONCEPT.166]: [2F_02.04.Zefiro-Europe.CONCEPT.167]: [2F_02.04.Zefiro-Europe.CONCEPT.170]: [2F_02.04.Zefiro-Europe.CONCEPT.182]: [2F_02.04.Zefiro-Europe.CONCEPT.324]: [2F_02.04.Zefiro-Europe.CONCEPT.325]: [2F_02.04.Zefiro-Europe.CONCEPT.326]: [2F_02.04.Zefiro-Europe.CONCEPT.332]: [2F_02.04.Zefiro-Europe.CONCEPT.333]: [2F_02.04.Zefiro-Europe.CONCEPT.335]: [2F_02.04.Zefiro-Europe.CONCEPT.338]: [2F_02.04.Zefiro-Europe.CONCEPT.340]: [2F_02.04.Zefiro-Europe.CONCEPT.66]: | Coupler cover control update | The chapter control of coupler cover has been detailed to express a more robust design in terms of availability of the train.<br>A logic was implemented to store the signals which are available for each flap.<br><br>A flap can be in two detectable positions. Open and Closed. And it can be locked in both. If a flap is in one of these positions the status is stored in TCMS<br><br>To ensure uninterrupted service in case of a single sensor failure the driver has the possibility to check the flaps manually and set the open/closed and locked stored signal manually.<br><br>As driving with open flaps is a safety relevant issue a failure event has been created based on the availability of the actual sensor signals and the stored signal.<br><br>Remove requirements 2F_02.04.Zefiro-Europe.CONCEPT.339 |
| 2F_02.04.ISSUES.-.Zefiro-Europe.24 | [2F_02.04.Zefiro-Europe.CONCEPT.353]: [2F_02.04.Zefiro-Europe.CONCEPT.354]: [2F_02.04.Zefiro-Europe.CONCEPT.355]: [2F_02.04.Zefiro-Europe.CONCEPT.356]: | Added requirements for incoherence check. | Added requirement for incoherence of el coupled signal (A and !A ; B and !B) |
| **Software baseline 0.5** | | | |
| 2F_02.04.ISSUES.-.Zefiro-Europe.25 | | Delete of double requirements for coupler control | Double Requirement. The same is stated in 2F_02.04.Zefiro-Europe.CONCEPT.98 2F_02.04.Zefiro-Europe.CONCEPT.99<br><br>CG 20130326:<br>Issue 748 on 4S TCMS SW |
| 2F_02.04.ISSUES.-.Zefiro-Europe.26 | [2F_02.04.Zefiro-Europe.CONCEPT.357]: | | A “flaps open and locked stored” signal shall be implemented based on the signals flaps open and flaps locked. Reset logic for this signal shall be implemented if the manual closure or automatic closure sequence is active.<br><br>CG 20130326:<br>Issue 749 on 4S TCMS SW<br><br>FD 20130415:<br>SIL added and issue closed |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.28 | [2F_02.04.Zefiro-Europe.CONCEPT.117]: | Prevent 2 coupler cover status signals to be active at the same time | It shall be prevented that the signals "flaps open and locked stored" and "flaps closed and locked stored" can be active at the same time.<br>Whenever one is selected, the other shall be reset.<br><br>FD 20130415:<br>req updated and issue closed |
| 2F_02.04.ISSUES.-.Zefiro-Europe.29 | [2F_02.04.Zefiro-Europe.CONCEPT.182]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.324]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.326]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.335]: | Improve functional performance of open and closure sequence | To improve the functional performance of the open and close sequences the signals "flaps open and locked stored" & "flaps closed and locked stored" shall be used instead of the single signals (open, locked, closed) for right and left flap.<br><br>CG 20130326:<br>Issue 752 on 4S TCMS SW<br><br>Remove requirements<br>2F_02.04.Zefiro-Europe.CONCEPT.327<br>2F_02.04.Zefiro-Europe.CONCEPT.329<br><br>Requirement 326 is not updated.<br><br>FD 20130415:<br>Issue closed |
| 2F_02.04.ISSUES.-.Zefiro-Europe.30 | [2F_02.04.Zefiro-Europe.CONCEPT.117]: | access definition of flap cover overwrite function | The function to overwrite the status of the coupler flaps position shall be possible for the front end rear end, and only available for the maintainer<br><br>CG 20130407:<br>Issue 766 on 4S TCMS SW |
| 2F_02.04.ISSUES.-.Zefiro-Europe.31 | [2F_02.04.Zefiro-Europe.CONCEPT.118]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.161]: | Update of max open and closure time requirement | There is no overall time requirement for the open sequence or closure sequence necessary. The requirements .118 and .161 describe events which shall be set if one of the 3 intermediate steps or each process fail. This is enough as it implies that the closure sequence or the open sequence failed.<br>(intermediate steps are:<br>unlocking, closing, locking or<br>unlocking, opening, locking)<br><br>CG 20130326:<br>Issue 760 on 4S TCMS SW<br><br>Remove requirements<br>2F_02.04.Zefiro-Europe.CONCEPT.98<br>2F_02.04.Zefiro-Europe.CONCEPT.99 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_A** | | |
|---|---|---|---|
| **Date:** | **2013-05-03** | | |
| **Editor:** | **G. Morgenstern** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.32 | [2F_02.04.Zefiro-Europe.CONCEPT.182]: [2F_02.04.Zefiro-Europe.CONCEPT.333]: [2F_02.04.Zefiro-Europe.CONCEPT.335]: | Removal of classification into automatic or manual closure | There shall be no difference between automatic and manual closure sequence. The sequence shall be triggered whenever the train speed is above 10 km/h and by the soft keys when it is ensured that the trains are separated after the decoupling<br><br>CG 20130326:<br>Issue 751 on 4S TCMS SW<br><br>Remove requirements<br>2F_02.04.Zefiro-Europe.CONCEPT.327<br>2F_02.04.Zefiro-Europe.CONCEPT.328<br>2F_02.04.Zefiro-Europe.CONCEPT.329 |
| 2F_02.04.ISSUES.-.Zefiro-Europe.33 | [2F_02.04.Zefiro-Europe.CONCEPT.338]: [2F_02.04.Zefiro-Europe.CONCEPT.357]: | Update of flaps open/closed and locked stored signals | The Signals "flaps open and locked stored" and "flaps closed and locked stored" shall be depending on all 6 signals of the flaps - open, closed, locked for left and right<br><br>CG 20130326:<br>Issue 750 on 4S TCMS SW<br><br>FD 20130415:<br>SIL added and issue closed |
| 2F_02.04.ISSUES.-.Zefiro-Europe.35 | [2F_02.04.Zefiro-Europe.CONCEPT.298]: [2F_02.04.Zefiro-Europe.CONCEPT.340]: | Adding TCMS coupler status signal | TCMS shall create a status signal for the train to identify it as coupled or not.<br><br>CG 20130407:<br>Issue 765 on 4S TCMS SW<br>Add SIL level<br><br>FD 20130415:<br>SIL added and issue closed |
| 2F_02.04.ISSUES.-.Zefiro-Europe.36 | | Old/ Double requirements removed | Old requirement to be removed. See also Issue 23<br><br>CG 20130407:<br>Issue 764 on 4S TCMS SW<br><br>Remove requirement<br>2F_02.04.Zefiro-Europe.CONCEPT.334 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

## 6.2 Revision log 3EST000225-7903_B

| **Revision:** | **_B** | | |
|---|---|---|---|
| **Date:** | **2018-05-15** | | |
| **Editor:** | **G. Parisi** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.37 | [2F_02.04.Zefiro-Europe.CONCEPT.315]: | | The current design of remote decoupling requires 50% parking brake presence to enable decoupling.<br>To ensure the safety of the train also on max inclination (40 per mille) tracks 75% is required.<br>The requirement needs to be specified to ensure that the requirement is applied for each train individually |
| 2F_02.04.ISSUES.-.Zefiro-Europe.38 | [2F_02.04.Zefiro-Europe.CONCEPT.301]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.302]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.303]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.304]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.305]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.306]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.318]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.319]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.321]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.360]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.365]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.366]: | Signal name clarification | Signal names and status have been clarified and updated according train electrical schematics<br><br>Mech Coupled Switch m1 → S14<br>Mech UNCoupled Switch m2 → S15<br><br>El Coupled relays<br>"Signal A" → R7a<br>El Coupled retained relays<br>"Signal B" → R9 |
| 2F_02.04.ISSUES.-.Zefiro-Europe.39 | [2F_02.04.Zefiro-Europe.CONCEPT.182]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.307]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.308]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.363]: | Definition of coupled/uncoupled | Train mechanical coupled status shall be defined based on both S14 and S15 signals<br><br>Both Switches shall state the same status otherwise TCMS shall set "unknown" and related functionalities shall be blocked<br>(eg remote decoupl, flap automatic closing)<br><br>Mechanical Coupling Status shall be allowed to change only when train speed is less than 3kph (in case of coupling status changes at speed - except if an unintended decoupling is detected, set a warning to the driver) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.40 | [2F_02.04.Zefiro-Europe.CONCEPT.207]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.309]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.364]: | Diagnosis of Train coupling and fault status | Updated events to diagnose train coupling status and fault conditions<br><br>(train mechanically coupled only status is not a fault) |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_B** | | |
|---|---|---|---|
| **Date:** | **2018-05-15** | | |
| **Editor:** | **G. Parisi** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.41 | [2F_02.04.Zefiro-Europe.CONCEPT.317]: | Decoupling ended | Condition speed >1km/h to define "train decoupled" to be checked<br><br>Train shall be defined as Decoupled when after a coupling condition NO mechanical nor electrical coupling is detected |
| 2F_02.04.ISSUES.-.Zefiro-Europe.42 | [2F_02.04.Zefiro-Europe.CONCEPT.315]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.316]: | Condition for remote decoupling update | Condition for remote decoupling<br>remove:<br>– "Line Voltage active"<br>– "Traction not blocked"<br><br>changed:<br>– "Forward direction" -> "Reverse direction NOT selected"<br>– train is coupled -> train mechanically coupled (mc) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.43 | [2F_02.04.Zefiro-Europe.CONCEPT.325]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.369]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.370]: | Front flaps sensor and valves | Added missing requirement to specify status and control signals |
| 2F_02.04.ISSUES.-.Zefiro-Europe.44 | [2F_02.04.Zefiro-Europe.CONCEPT.182]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.338]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.357]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.371]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.373]: | Front Flap Status definition | To avoid false signalling of cover position (due to single sensor fault). New logic has been introduced to define open/unlocked status based on signals coming from open/closed and locked/unlocked sensors<br><br>In case of fault cover<br>automatic closing and remote closing will be inhibited |
| 2F_02.04.ISSUES.-.Zefiro-Europe.45 | [2F_02.04.Zefiro-Europe.CONCEPT.117]: | Flap position override | Flap position override will be allowed for driver (not only maintainer) on the IDU |
| 2F_02.04.ISSUES.-.Zefiro-Europe.46 | [2F_02.04.Zefiro-Europe.CONCEPT.119]: | Cover Closing command interlock | Any Cover closing command shall be inhibited where Automatic coupler status is detected as Mechanically Coupled (mc) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.47 | [2F_02.04.Zefiro-Europe.CONCEPT.298]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.346]: | Speed Limitation due to coupler covers open | Speed Limit will be increased from 140km/h to 250km/h<br>(project decision)<br><br>New condition to set limitation according updated flaps and automatic coupler status definition<br><br>Speed limit will be applied only in case front covers are open (and will be bypassed in case flap position is overridden by IDU) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.48 | [2F_02.04.Zefiro-Europe.CONCEPT.374]:<br>[2F_02.04.Zefiro-Europe.CONCEPT.375]: | Fault detection of de coupling valves | Added event to detects incoherence on decoupling valves |
| 2F_02.04.ISSUES.-.Zefiro-Europe.51 | [2F_02.04.Zefiro-Europe.CONCEPT.201]: | Coupling status | Removed requirement 201 since already covered by concept.363 |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.

| **Revision:** | **_B** | | |
|---|---|---|---|
| **Date:** | **2018-05-15** | | |
| **Editor:** | **G. Parisi** | | |
| **Change ID DOORS** | **Refers to requirement ID** | **Title** | **Change content** |
| 2F_02.04.ISSUES.-.Zefiro-Europe.52 | [2F_02.04.Zefiro-Europe.CONCEPT.371]: [2F_02.04.Zefiro-Europe.CONCEPT.373]: | Cover remote command | Removed inhibition of cover remote closing / opening command in case of cover faulty<br><br>Increased time from 5s to 15s<br><br>(after type test result) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.53 | [2F_02.04.Zefiro-Europe.CONCEPT.340]: | Event for cover closing when train is coupled | Removed event already covered from diagnostic |
| 2F_02.04.ISSUES.-.Zefiro-Europe.54 | [2F_02.04.Zefiro-Europe.CONCEPT.184]: [2F_02.04.Zefiro-Europe.CONCEPT.324]: [2F_02.04.Zefiro-Europe.CONCEPT.326]: [2F_02.04.Zefiro-Europe.CONCEPT.335]: [2F_02.04.Zefiro-Europe.CONCEPT.376]: [2F_02.04.Zefiro-Europe.CONCEPT.384]: [2F_02.04.Zefiro-Europe.CONCEPT.385]: [2F_02.04.Zefiro-Europe.CONCEPT.386]: [2F_02.04.Zefiro-Europe.CONCEPT.387]: | Front flap buzzer introduction | Control of acoustic buzzer 10s before open/closure sequence activation by IDU |
| 2F_02.04.ISSUES.-.Zefiro-Europe.55 | [2F_02.04.Zefiro-Europe.CONCEPT.171]: [2F_02.04.Zefiro-Europe.CONCEPT.315]: [2F_02.04.Zefiro-Europe.CONCEPT.320]: [2F_02.04.Zefiro-Europe.CONCEPT.388]: | Local decoupling via TCMS | Today local decoupling has been realized by a wired circuit. This circuit doesn't take in consideration if the trains are braked.<br>TCMS does that but only for remote decoupling.<br>For this reason the cheaper modification for guarantee the function safety is to move the local command under TCMS scope.<br>Proposed modification: Cut out the wiring commands for local decoupling and implement inside TCMS SW the local command decoupling (same concept used for remote decoupling) |
| 2F_02.04.ISSUES.-.Zefiro-Europe.57 | [2F_02.04.Zefiro-Europe.CONCEPT.315]: | Decoupling via TCMS without train coupled pre-condition | Decoupling command is possible only if the train is coupled. It depends of the coupler microswitch status (mechanical coupling status).<br>The proposal consists of removing this interlock (HW and SW) for improving availability of this function. |
| 2F_02.04.ISSUES.-.Zefiro-Europe.58 | [2F_02.04.Zefiro-Europe.CONCEPT.376]: [2F_02.04.Zefiro-Europe.CONCEPT.384]: | | buzzer duration lowered from 10 sec to 5 sec - decision at FCCB |


This document and its contents are the property of Bombardier Inc. or its subsidiaries. This document contains confidential proprietary information. The reproduction, distribution, utilization or the communication of this document or any part thereof, without express authorization is strictly prohibited. Offenders will be held liable for the payment of damages.

©2018, Bombardier Inc. or its subsidiaries. All rights reserved.